More on Transfer Functions

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1 Introduction

In this Lecture we go back to the properties of the transfer function first listed in Part 4, and develop each one. The sections of this lecture are:

1. Components of a Transfer Function

   (a) All information about the signals is included in the Laplace Transforms of the input and output signals

   (b) All information about the system is included in the transfer function $H(s)$!

2. The transfer function has units of gain

3. Pole-placement design

4. Conclusions

This lecture relates to Franklin *et al.* section 3.2 (7th ed.).

2 Components of a Transfer Function

- All information about the signals is included in the Laplace Transforms of the input and output signals.

- All information about the system is included in the transfer function $G_{uy}(s)$!

- The transfer function is an unusual species: it is a ratio of Laplace Transforms (of signals), yet it captures only information about the system.

  - The controls engineer rarely, if ever, knows the specific signals a controller will encounter in use, and so can not calculate the $U(s)$ or $Y(s)$ of actual operation.

  - We can assume ideal inputs, such as a step input, $U(s) = 1/s$, and determine $Y(s)$ and all other signals in the system.

  - The transfer function is the generalized gain, it has the ratio of Output over Input Laplace transforms (and has the units of the [output/input]).

$$G_{uy}(s) = \frac{Y(s)}{U(s)} \quad ; \quad \frac{\text{LT of Output}}{\text{LT of Input}} \quad ; \quad \left[\frac{\text{Units of Output}}{\text{Units of Input}}\right] \tag{1}$$

  - Choosing $s = j\omega$

$$G_{uy}(s)\Big|_{s=j\omega}$$

    gives the generalized gain of $G_{uy}(s)$ at frequency ω .

2.1 The transfer function is linked to the governing differential equation

- In general, for the TF:

$$G_p(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0} \tag{2}$$

Then the governing differential equation is:

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \cdots a_1 \dot{y}(t) + a_0 y(t) \tag{3}$$

$$= \quad b_m \frac{d^m u(t)}{dt^m} \cdots + b_1 \dot{u}(t) + b_0 u(t)$$

2.1.1 TF $\longleftrightarrow$ Diff Eq Example (review)

- Based on Eqns (2) and (3), we can just cut-and-paste coefficients from the the Diff Eq to the TF, or vice versa. For example, if

$$H(s) = \frac{3s^2 + 18s + 24}{s^3 + 4s^2 + 3s + 0} \tag{4}$$

then:

$$1 \frac{d^3 y(t)}{dt^3} + 4 \ddot{y}(t) + 3 \dot{y}(t) + 0 y(t) = 3 \ddot{u}(t) + 18 \dot{u}(t) + 24 u(t) \tag{5}$$

3 The transfer function has units of gain

- One interesting aspect of $H(s)$ is that it is the generalized gain of the system.

3.1 DC gain: the limit as $s \to 0$ of $H(s)$

- The DC gain of a system is the value of its steady out for a steady input.
- For example:
  - Given $H(s) = \dfrac{3s+40}{2s^2+12s+25}$, the DC gain is

$$K_{DC} = \lim_{s\to 0} H(s) = \frac{3\cdot 0+40}{2\cdot 0+12\cdot 0+25} = \frac{40}{25} = 1.625$$

    - The DC gain is determined by the s^0 coefficients of the polynomials.
    - Compare with K_{rlg} and K_{Bode}.

  - Given $H(s) = \dfrac{3s^2+18s+24}{s^3+4s^2+3s+0}$, the DC gain is:

$$K_{DC} = \lim_{s\to 0} H(s) = \frac{3\cdot 0+18\cdot 0+24}{0+4\cdot 0+3\cdot 0+0} = \frac{24}{0} = \infty$$

$\frac{1}{s}$ Gives infinite DC gain

  - Systems with a pole at the origin have infinite DC gain.
  - An infinite gain system may seem surprising, but we often engineer systems to have infinite DC gain, to make steady-state errors go to zero !

DC gain (continued)

- Consider that, as seen in figure 1, $H(s)$ factors into:

$$H(s) = \frac{1}{s} \frac{3s^2 + 18s + 24}{s^2 + 4s + 3} \tag{6}$$

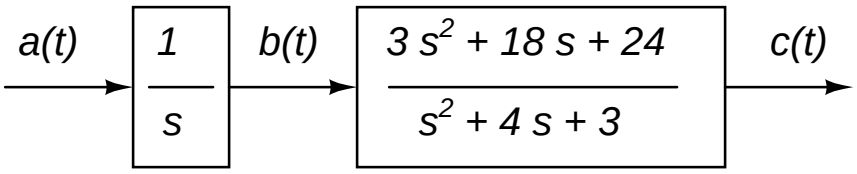


Figure 1: A system with an integrator.

  - The DC gain from $b(t)$ to $c(t)$ is 8.
  - However, with the transfer function from $a(t)$ to $b(t)$ is

$$\frac{B(s)}{A(s)} = \frac{1}{s}$$

    which corresponds to:

$$b(t) = \int_{\tau=0}^{t} a(\tau)\, d\tau$$

- And so, if $a(t)$ is a constant (DC) input, $b(t)$ goes to infinity.
- A pole at the origin corresponds to an integrator.
  - The DC gain of an integrator is ∞ .
  - Integrators are common in systems of interest for controls, for example:
    * Position is the integral of velocity. For mechanisms or vehicles, the transfer function from $\dot{x}(t)$ to $x(t)$ has infinite DC gain.
    * If the system keeps going at constant $\dot{x}$, in infinite time it goes infinitely far.

3.2 The generalized (or complex) gain applies to sine-wave inputs

- For a linear system, if input $u(t)$ is a sinusoid, then output $y(t)$ is a sinusoid of the same frequency.

- Using $s = j\omega$, the Transfer Function gives the complex gain at frequency ω. Use $s = j\omega$

  – For a given ω,

$$H(j\omega) = M(\omega)\, e^{j\phi(\omega)} = M(j\omega) < \phi(j\omega) \tag{7}$$

    * $M(w)$: Amplitude Gain [Units of gain]
    * $\phi(w)$: Phase shift [degrees]

  – We can represent both $u(t)$ and $y(t)$ as phasors:

$$u(t) = M_u \cos(\omega t + \phi_u) \quad \longleftrightarrow \quad \vec{u} = M_u \angle \phi_u$$

  and

$$y(t) = M_y \cos(\omega t + \phi_y) \quad \longleftrightarrow \quad \vec{y} = M_y \angle \phi_y$$

  where M_u and ϕ_u are the magnitude and phase of $u(t)$, and M_y and ϕ_y are the magnitude and phase of $y(t)$.

- Now, just as

$$Y(s) = H(s)\, U(s) \tag{8}$$

  we have:

$$\vec{y} = H(j\omega)\, \vec{u} \tag{9}$$

  And so:

$$H(j\omega) = \left.\frac{\vec{y}}{\vec{u}}\right|_{\text{given } \omega} \tag{10}$$

3.3 Example: $H(j\omega)$ as complex gain

- Using:

$$H(s) = \frac{3s^2 + 18s + 24}{s^3 + 4s^2 + 3s + 0} \tag{11}$$

with $u(t)$ given as:

$$u(t) = 5.0\cos(20t + 30^o) \tag{12}$$

Find $y(t)$.

1. To represent $u(t)$ as a phasor, we find from Eqn (12) that:

$$\vec{u} = 5.0\angle 30^o \qquad (\text{with} \quad \omega = 20\ [\text{rad/sec}]) \tag{13}$$

2. Find the complex gain of $H(s)$ at $\omega = 20$ [rad/sec]. Plugging in $s = j20$:

$$\begin{aligned} H(s)\Big|_{s=j20} &= \frac{3(j20)^2 + 18(j20) + 24}{(j20)^3 + 4(j20)^2 + 3(j20) + 0} = \frac{-1176 + 360j}{-1600 - 7940j} \\ &= \frac{1230.0\angle 162.98^o}{8099.60\angle -101.39^o} = 0.152\angle -95.63^o \end{aligned}$$

3. Finally, determine $\vec{y}$:

$$\begin{aligned} \vec{y} = H(j\omega)\,\vec{u} &= (0.152\angle -95.63^o)\,(5.0\angle 30^o) \\ &= 0.759\angle -65.63^o \end{aligned}$$

which gives the output corresponding to $u(t)$:

$$y(t) = 0.759\cos(20t - 65.63^o) \tag{14}$$

3.4 Plotting $H(j\omega)$

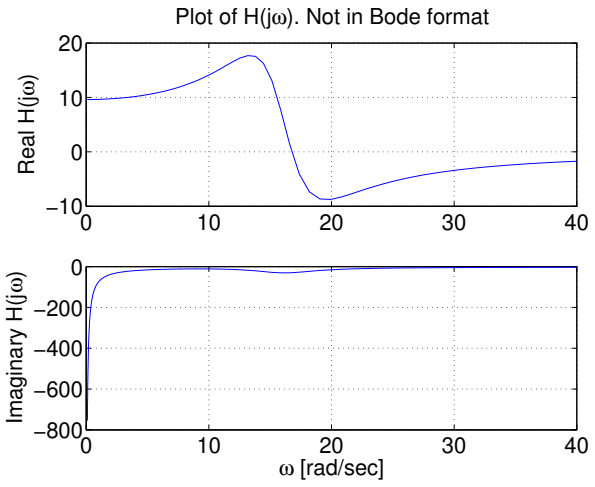
- With

$$H(j\omega) = M(\omega) \angle \phi(\omega)$$

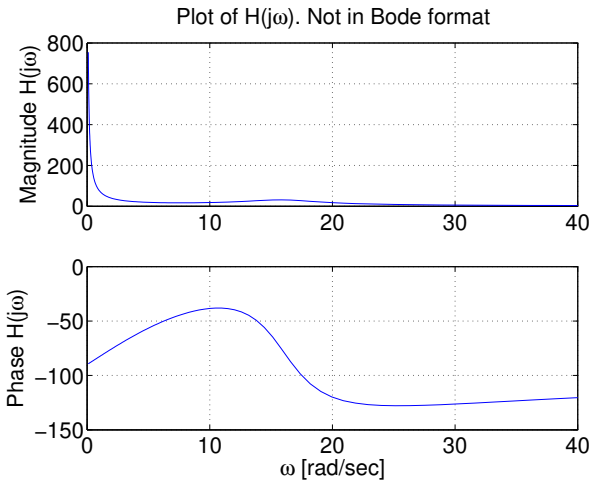
$M(\omega)$ and $\phi(\omega)$ can be plotted as a function of ω. For example, using

$$H(s)\Big|_{s=j\omega} = \frac{100\left(s^2+30s+200\right)}{s^3+6s^2+265s+0} = \frac{100\left((j\omega)^2+30(j\omega)+200\right)}{(j\omega)^3+6(j\omega)^2+265(j\omega)+0}$$

- We could plot
  - (A) the real and imaginary parts of $H(j\omega)$ vs. ω, or
  - (B) the magnitude and phase of $H(j\omega)$ vs. ω.
- However, figures 2(A) and 2(B) are not very informative.



A) Plot of real and imaginary parts of $H(j\omega)$.



B) Plot of magnitude and phase of $H(j\omega)$

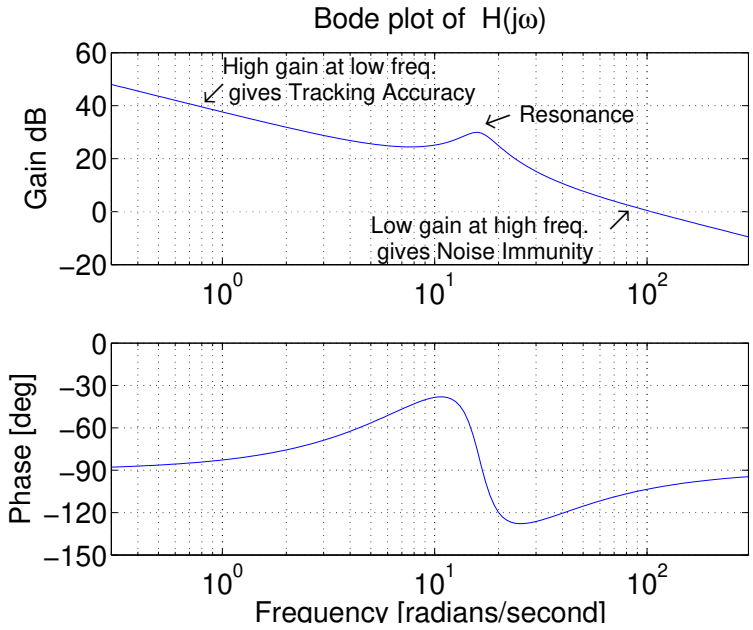
Figure 2: Possible plots of $H(j\omega)$. These carry information about the transfer function, but are not the Bode plot.

3.4.1 Bode plot

- In the 1920's H. W. Bode, of Bell Labs, worked out a graphical representation of $H(j\omega)$ that is ***very*** informative of system behavior.
  - It came to be known as the Bode plot, and has been long and widely used for control system and filter design.
  - Starting with

$$H(s)\Big|_{s=j\omega} = M(\omega)\angle\phi(\omega) \tag{15}$$

    1. Make the ω axis logarithmic, base 10.
    2. Magnitude plot in dB: $M_{dB}(\omega) = 20\log_{10}(M(\omega))$.
    3. Plot phase in degrees.



$M_{dB}(\omega)$

$\phi(\omega)$

- Many properties of a control system can be determined by interpreting the Bode plot of the loop gain, including:
  - Bandwidth
  - Tracking Accuracy
  - Phase Margin
  - Noise Immunity

(Analysis of the Bode plot is a topic of EE 574).

4 Conclusions

- In figure 3, inputs and outputs are connected in two ways:
  - Via the differential equation in the time domain, and
  - Via the transfer function in the Laplace domain.

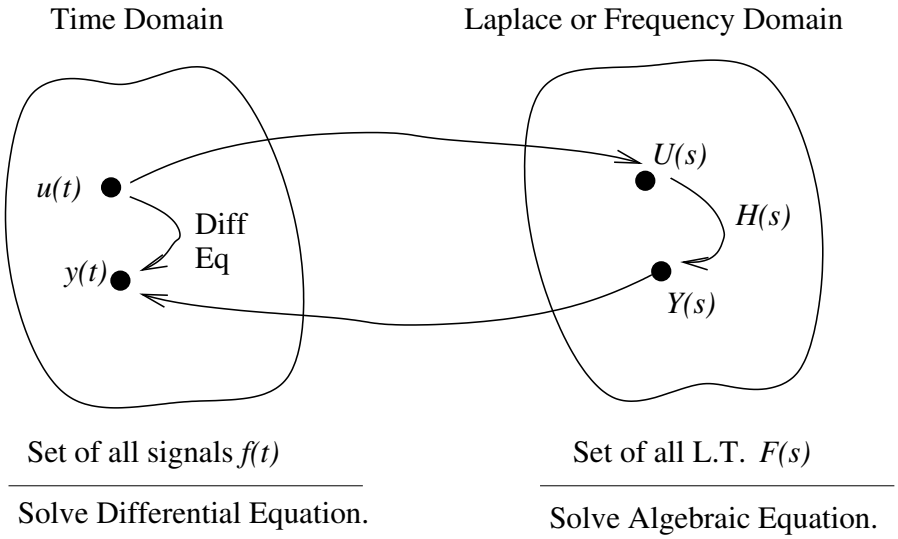


Figure 3: Mapping from $u(t)$ to $y(t)$ by Laplace Transform $H(s)$.

- A Diff Eq can be solved using the TF, which involves traversing the entire loop, from $u(t)$ to $y(s)$.
- Alternatively, we can say many things about the system by considering only $H(s)$ (such as T_r, T_s, Percent Overshoot, etc.).

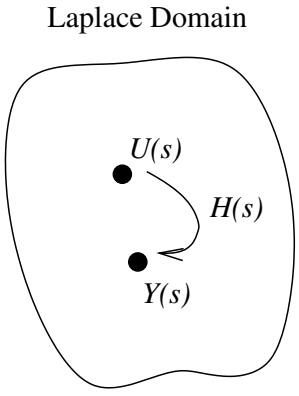


Figure 4: Laplace Transform $H(s)$.

- This way we are never concerned with solving for any signal !